Dhruv Nirmal

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PROFESSIONAL SUMMARY

I'm a data scientist who pairs predictive modelling with hands-on GenAI and agentic AI, building both into systems that support live decisions rather than one-off demos. I led a one-vs-all classification model that brought structure to roughly A$12B of client spend, and built an agentic reporting workflow with an independent verification agent that re-checks the numbers before a human signs off, my way of making sure AI output earns trust rather than assumes it. I also build Prophet forecasting models and ran a pricing analysis that helped a client find around A$2M in savings, always translating the results into language non-technical stakeholders can act on. I work mainly in Python and SQL, and like owning a problem end to end, from framing it with the client through to the model, or the agent, running in production.

KEY SKILLS

Predictive Modelling: scikit-learn, classification (logistic regression, TF-IDF, embeddings), Prophet time-series forecasting, feature engineering, cross-validation, hyperparameter tuning (GridSearchCV), class-imbalance handling; familiar with gradient boosting (GBM), keen to build on it in production

GenAI & Agentic AI: LLM-driven workflows (Claude, ChatGPT), prompt design, multi-agent orchestration, independent QA/verification agents, structured tool calls, evaluating and refining AI outputs before they reach a stakeholder; familiar with retrieval-augmented generation (RAG)

Python & Data: Python (pandas, numpy, scikit-learn), SQL, PySpark and Databricks (internship), data pipelines, feature engineering

Cloud: Azure (Microsoft Graph API ingestion, work), AWS (S3, EC2, IAM, personal build with Terraform and Snowflake)

Visualisation & Communication: Tableau, Power BI, translating model and AI outputs into plain language for non-technical stakeholders

EXPERIENCE

Data Scientist, Purchasing Index Data Analytics (Comprara Group)

Jun 2024 to Present | Melbourne, Australia

A data and analytics consultancy. I own client engagements end to end: framing the problem, building the model or the agent, and presenting the results to commercial stakeholders. Selected engagements:

Agentic client reporting with an independent QA agent, internal, my own firm

- Problem: our analysts were each spending up to 30 minutes per client, per reporting cycle, hand-assembling progress reports across roughly 25 to 30 client accounts company-wide.
- Approach: built an agentic workflow (Claude, Python) that drafts the report from the underlying analytics outputs, then added a second, independent verification agent, run with no memory of the drafting session, that re-derives the key numbers from source data and checks them against the draft before a human reviews and sends it.
- Result: rolled out company-wide, saving roughly 12 to 15 hours a week across the team, with a human check on every output before it reaches a client, standardised and trustworthy AI-assisted delivery rather than a one-off automation.

ML spend classification, enterprise procurement client (~A$12B spend)

- Problem: only around 60 to 65% of a client's five-year, roughly A$12B spend book was reliably categorised; the manual review to close the gap took an account manager over a month per cycle.

- Approach: led the firm's first ML classifier for this problem, a one-vs-all logistic regression (one model per category), with features engineered from spend value, line count and transaction text (TF-IDF, moving toward embeddings), tuned with GridSearchCV and cross-validation, with explicit handling for the heavy class imbalance; kept the high-value head of spend on manual review and modelled the long tail.
- Validation: evaluated on precision, recall and F1 given the imbalance; the model is retrained as new spend data comes in.
- Result: cut the categorisation cycle from over a month to a single day's model run; the pipeline is documented and now reused across other clients.

Pricing and cost analytics, multi-venue restaurant group (AU/NZ)

- Problem: the client felt their supply costs were too high but couldn't see where the money was leaking across products, venues and suppliers.
- Approach: owned the account as the sole analyst, built a product catalogue of roughly 3,000 items, then a pricing-decision dashboard (Python, SQL) tracking price movement and cost leakage; presented findings directly to the Chief Procurement Officer and category managers, translating the model outputs into plain, relatable terms for non-technical stakeholders.
- Result: helped the client identify close to 30% savings, roughly A$2M over a year, in their largest category, used directly in supplier negotiations.

Demand forecasting, fresh-produce and agricultural client

- Problem: the client needed to plan raw-material and chemical inventory three months ahead, with chemicals sourced internationally and exposed to shipping cost swings.
- Approach: built a Prophet time-series model with external regressors (sea-freight trends, input prices), prepared the data with pandas, and delivered the forecast through a planning dashboard.
- Result: forecasts within a 12.5 to 14% error margin, comfortably inside the client's tolerance, giving the client a three-month forward view for ordering.

Data Engineer Intern, Victorian Centre for Data Insights (VCDI)

Aug 2023 to Nov 2023 | Melbourne, Australia

- Built a distributed anomaly-detection pipeline in Databricks using PySpark on government procurement data, engineering scalable transformation layers, improving detection accuracy by around 20%.
- Delivered findings through a Power BI solution adopted by senior Department of Transport stakeholders, presenting system architecture and model outcomes to a cross-functional audience.

PROJECTS

Job Hunt OS: multi-agent agentic AI system (Claude Code, Python)

- Designed and built a file-based, multi-agent career-search system: an orchestrator agent that delegates to specialist agents (resume tailoring, outreach, career coaching), running on scheduled cloud routines with a human review gate before anything is sent to a real person. A personal build that puts agentic system design, prompt engineering and multi-agent orchestration into daily practice outside of client work.

Cloud Data Platform with Terraform (personal build)

- Provisioned cloud infrastructure as code on AWS (S3, EC2, IAM) with Terraform, with an automated ingestion pipeline into Snowflake and a Power BI layer on top, modular across dev/staging/prod environments.

EDUCATION

Master of Data Science, Monash University | Feb 2022 to Dec 2023

Bachelor of Engineering, Thapar University | May 2017 to May 2021